

CAEN CARPIQUET, France

WMO: 070270

Lat: 49.1800N

Lon: 0.4561W

Elev: 256

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	28.5	20.0	15.1	28.4	23.1	17.6	30.1	33.6	50.2	30.6	50.6	6.6	130	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.4	82.4	67.6	78.3	66.0	74.9	64.6	69.2	79.8	67.2	75.9	65.5	72.9	9.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.6	73.5	63.9	89.8	71.3	62.5	85.3	69.4	33.4	80.3	31.8	75.9	30.5	72.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	23.2	20.5	20.8	91.2	4.5	4.6	17.5	94.5	14.9	97.3	12.3	99.8	9.0	103.2
			19.8	73.1	4.4	2.1	16.6	74.6	14.0	75.8	11.5	76.9	8.3	78.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	41.8	42.6	45.9	49.4	54.9	60.2	64.1	64.2	59.8	54.6	47.5	42.7	(6)
	DBStd	9.71	6.71	5.98	5.35	5.14	5.03	4.93	4.41	4.67	4.57	5.42	5.47	6.71	(7)
	HDD50	1067	259	212	148	71	12	0	0	0	1	21	107	236	(8)
	HDD65	4726	719	627	592	467	315	161	71	70	166	323	525	690	(9)
	CDD50	1934	6	5	21	53	164	305	436	441	296	163	33	11	(10)
	CDD65	118	0	0	0	0	2	16	41	47	11	1	0	0	(11)
	CDH74	982	0	0	0	4	31	182	319	347	90	9	0	0	(12)
	CDH80	256	0	0	0	0	3	55	76	104	17	1	0	0	(13)
(14) Wind	WSAvg	9.8	11.8	11.4	10.6	9.6	9.0	8.4	8.7	8.6	8.8	9.7	10.5	11.2	(14)
(15) Precipitation	PrecAvg	30.0	2.7	2.3	2.0	2.2	2.4	2.2	2.2	2.3	2.2	3.0	3.2	3.1	(15)
	PrecMax	39.5	6.6	5.3	6.6	4.6	4.6	6.3	5.3	5.2	5.0	6.4	5.7	7.8	(16)
	PrecMin	20.2	0.4	0.4	0.3	0.2	0.5	0.4	0.4	0.4	0.3	0.2	0.8	0.4	(17)
	PrecStd	4.3	1.4	1.3	1.3	1.2	1.1	1.3	1.2	1.5	1.1	1.4	1.3	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	59.5	65.7	73.3	78.0	87.1	87.6	89.3	82.2	74.1	62.9	57.9	(19)
		MCWB	53.7	51.8	54.7	60.5	64.8	69.6	69.0	69.2	65.8	63.9	59.0	54.8	(20)
	2%	DB	55.1	55.4	60.6	66.5	72.2	79.0	80.9	82.1	75.9	68.2	59.5	56.2	(21)
		MCWB	52.4	51.1	53.4	56.6	61.9	66.4	67.3	67.0	64.2	61.6	56.2	53.5	(22)
	5%	DB	53.0	53.3	57.4	62.2	68.3	73.7	77.1	77.2	71.6	65.0	57.6	54.4	(23)
		MCWB	50.5	49.8	52.2	54.4	60.2	64.4	65.6	65.9	62.6	59.8	54.6	52.0	(24)
	10%	DB	51.0	51.5	54.6	58.6	64.7	70.0	73.7	73.5	68.4	62.7	55.6	52.4	(25)
		MCWB	48.6	48.5	50.4	52.1	57.8	62.0	64.0	63.9	60.8	58.4	52.8	50.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	53.6	56.8	61.1	66.6	71.7	71.3	71.5	68.3	64.8	59.9	55.8	(27)
		MCDB	56.2	57.2	62.9	70.1	75.3	83.2	84.0	84.2	77.4	71.0	62.2	57.4	(28)
	2%	WB	52.8	51.8	54.6	57.6	63.2	67.8	68.6	69.0	65.6	62.6	56.9	53.9	(29)
		MCDB	54.7	54.6	59.0	65.3	70.5	77.4	78.6	79.5	73.2	67.2	59.2	55.8	(30)
	5%	WB	50.7	50.3	52.7	55.2	60.6	65.0	66.6	67.0	63.6	60.6	54.9	52.3	(31)
		MCDB	52.8	53.0	56.6	60.8	67.0	72.3	74.8	75.1	69.8	64.1	57.3	54.3	(32)
	10%	WB	48.8	48.6	50.9	53.0	58.5	62.8	65.0	65.1	62.0	59.1	53.2	50.3	(33)
		MCDB	50.7	51.2	54.3	57.6	63.9	68.8	72.1	71.6	67.2	62.1	55.3	52.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	11.0	13.2	15.5	15.8	16.9	17.5	17.4	16.4	13.4	10.7	9.6	(35)
		MCDBR	10.0	13.4	18.1	22.1	22.5	24.1	24.1	23.7	21.7	16.4	11.8	9.8	(36)
	5% WB	MCWBR	8.9	10.1	12.0	13.6	13.6	13.7	11.9	11.2	11.8	10.4	9.3	8.7	(37)
		MCWBR	9.9	11.9	15.9	19.4	20.4	22.2	21.4	21.0	19.0	14.7	10.8	9.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.326	0.343	0.376	0.394	0.400	0.399	0.393	0.387	0.371	0.357	0.333	0.320	(40)
		taud	2.425	2.392	2.295	2.260	2.284	2.322	2.341	2.379	2.407	2.441	2.443	2.394	(41)
	Ebn at Noon		230	254	262	270	273	273	273	270	262	245	225	214	(42)
		Edh at Noon	21	27	34	39	40	39	38	35	30	25	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		290	527	896	1395	1647	1832	1762	1479	1144	648	354	239	(44)
	RadStd		27	63	99	132	133	184	116	108	77	53	41	30	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page